

TEAM BEAM

# FSD-Project

## Classifying Deceleration Measurement Sequences

Ben Krüger, Erik Wiese, Anna Kugelmann, Martin Wellendorf  
Monday, 23.06.2025

# Contents

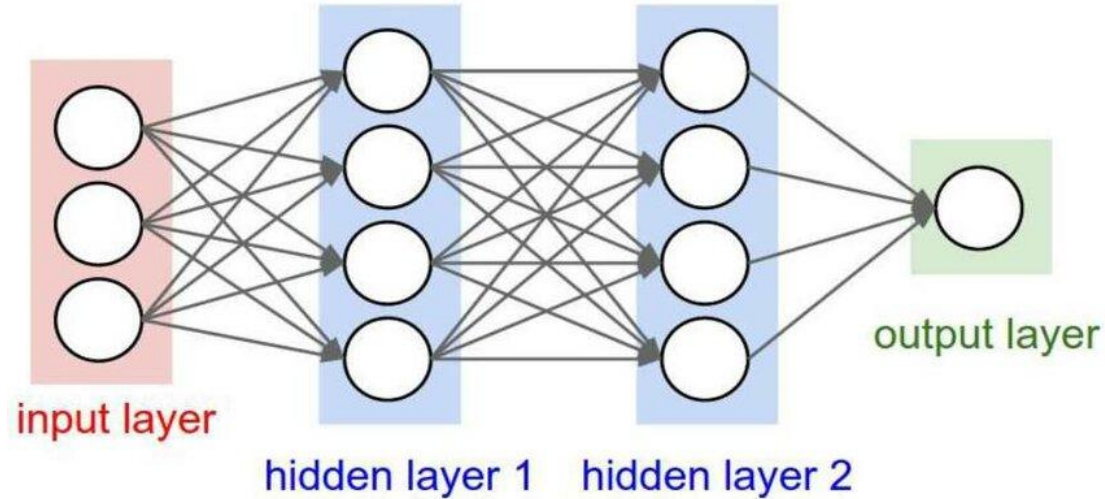
- **Task 1**
  - Hyperparameter Optimizer
  - Fully Connected Network
  - Convolutional Network
  - Long Short-Term Memory Network
  - Comparison
- **Task 2**
  - Analysis with Respect to Metadata
  - Data Preprocessing
  - Data Augmentation
  - Debiasing Autoencoder
- **Summary**

# Hyperparameter Optimizer

| hyperparameter | type        | range                                         | explanation                                                                                                |
|----------------|-------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------|
| batch size     | integer     | 16, 32, 64, 128, 256                          | Number of data samples that are processed simultaneously within one training step.                         |
| learning rate  | float       | $[10^{-5}, 10^{-2}]$                          | Learning rate of the optimizer used in training.                                                           |
| dropout        | float       | $[0, 0.5]$                                    | Proportion of neurons that will be randomly selected and whose output will be ignored (i.e., set to zero). |
| optimizer      | categorical | { adadelta, adam, adamw, lion, rmsprop, sgd } | Optimizer to be used in training.                                                                          |

Table 1: Space of architecture-independent hyperparameters

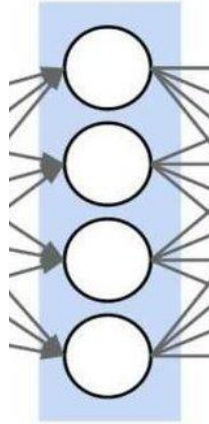
# Fully Connected Network



| hyperparameter         | type    | range                    | explanation                               |
|------------------------|---------|--------------------------|-------------------------------------------|
| number of Dense layers | integer | $[1, 6] \cap \mathbb{Z}$ | Number of Dense layers used in the model. |

Automatic Fruit Detection System using Multilayer Deep Convolution Neural Network - Scientific Figure on ResearchGate. Available from: [https://www.researchgate.net/figure/Fully-Connected-Network\\_fig1\\_352393254](https://www.researchgate.net/figure/Fully-Connected-Network_fig1_352393254) [accessed 20 Jun 2025]

# Fully Connected Network

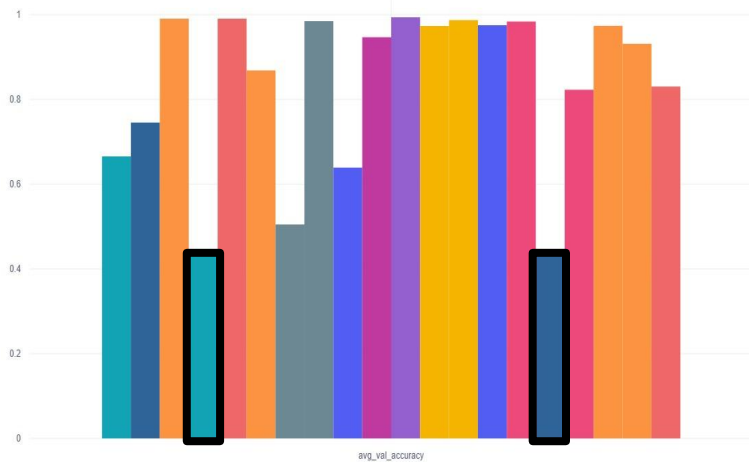


hidden layer 1

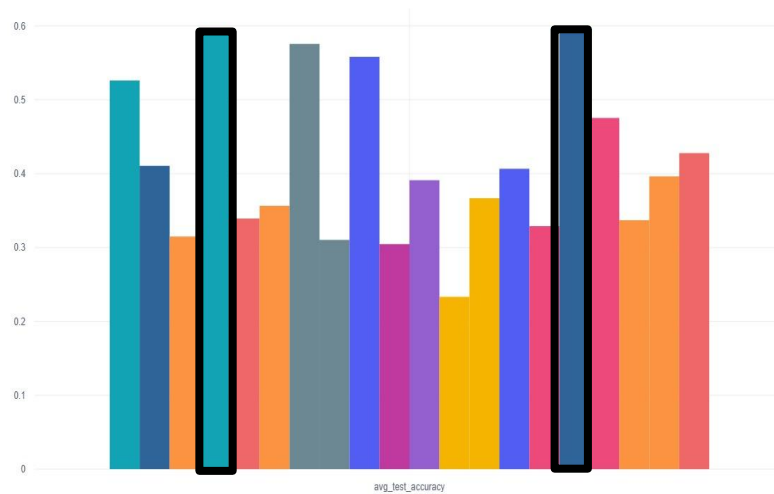
| hyperparameter         | type    | range                                    | explanation                                   |
|------------------------|---------|------------------------------------------|-----------------------------------------------|
| number of Dense layers | integer | $[1, 6] \cap \mathbb{Z}$                 | Number of Dense layers used in the model.     |
| units                  | integer | $\{32, 64, 128, 256\}$                   | Number of output features of the Dense layer. |
| activation             | integer | $\{\tanh, \text{sigmoid}, \text{relu}\}$ | Activation function for the output.           |

Automatic Fruit Detection System using Multilayer Deep Convolution Neural Network - Scientific Figure on ResearchGate. Available from: [https://www.researchgate.net/figure/Fully-Connected-Network\\_fig1\\_352393254](https://www.researchgate.net/figure/Fully-Connected-Network_fig1_352393254) [accessed 20 Jun 2025]

# Fully Connected Network

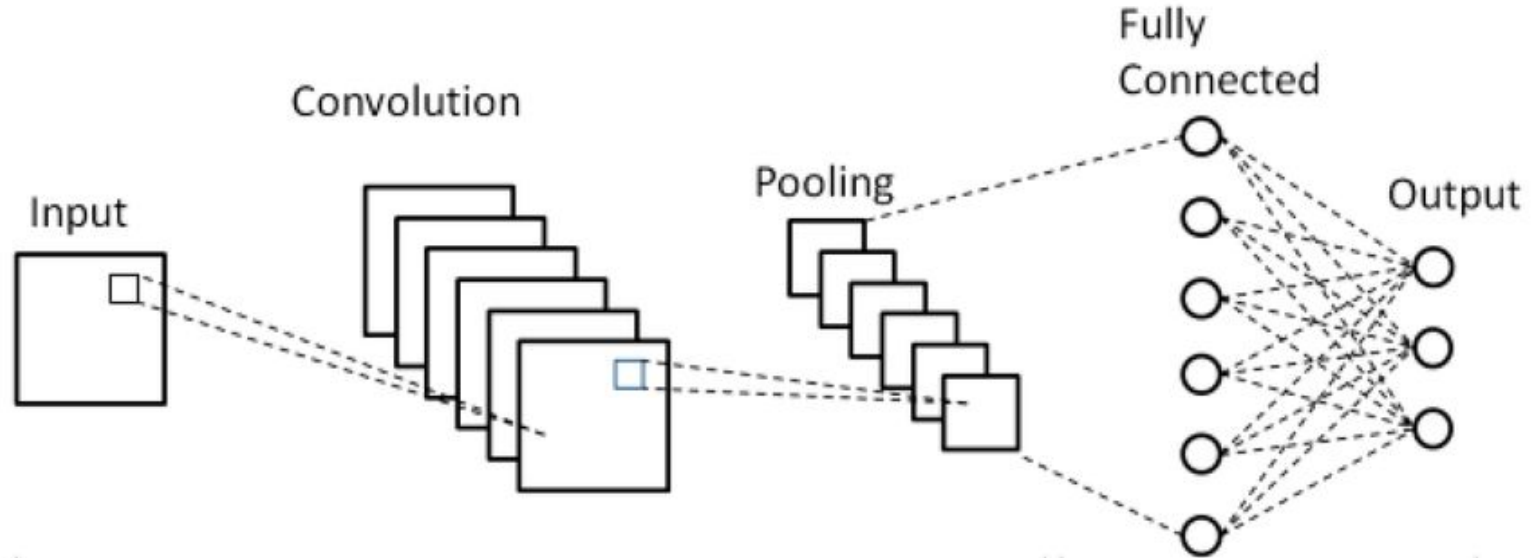


43,67%



59,24%

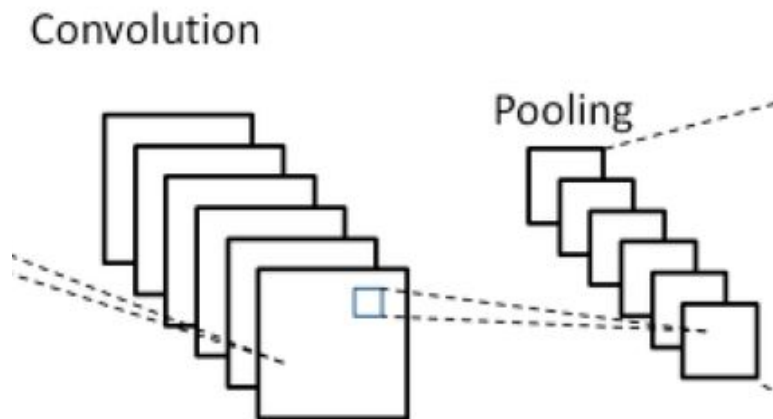
# Convolutional Network



| hyperparameter         | type    | range                    | explanation                                       |
|------------------------|---------|--------------------------|---------------------------------------------------|
| number of Conv layers  | integer | $[1, 3] \cap \mathbb{Z}$ | Number of convolutional layers used in the model. |
| number of Dense layers | integer | $[1, 3] \cap \mathbb{Z}$ | Number of Dense layers used in the model.         |

[https://www.researchgate.net/figure/Schematic-diagram-of-a-basic-convolutional-neural-network-CNN-architecture-26\\_fig1\\_336805909](https://www.researchgate.net/figure/Schematic-diagram-of-a-basic-convolutional-neural-network-CNN-architecture-26_fig1_336805909)[accessed 21 June 2025]

# Convolutional Network

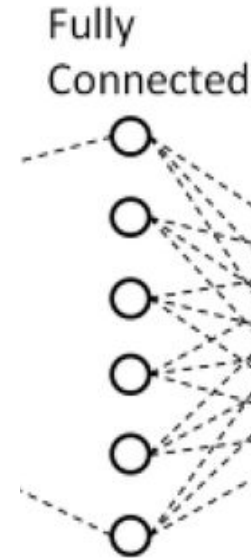


| hyperparameter  | type        | range                        | explanation                                         |
|-----------------|-------------|------------------------------|-----------------------------------------------------|
| filters         | integer     | {16, 32, 64, 128}            | Number of filters of the convolutional layer.       |
| conv activation | categorical | {tanh, sigmoid, relu}        | Activation function.                                |
| kernel size     | categorical | {(3,1), (3,2), (3,3), (5,5)} | Size of the kernel used in the convolutional layer. |
| pooling         | categorical | {avg, max}                   | Type of pooling used in the model.                  |

[https://www.researchgate.net/figure/Schematic-diagram-of-a-basic-convolutional-neural-network-CNN-architecture-26\\_fig1\\_336805909](https://www.researchgate.net/figure/Schematic-diagram-of-a-basic-convolutional-neural-network-CNN-architecture-26_fig1_336805909)[accessed 21 June 2025]



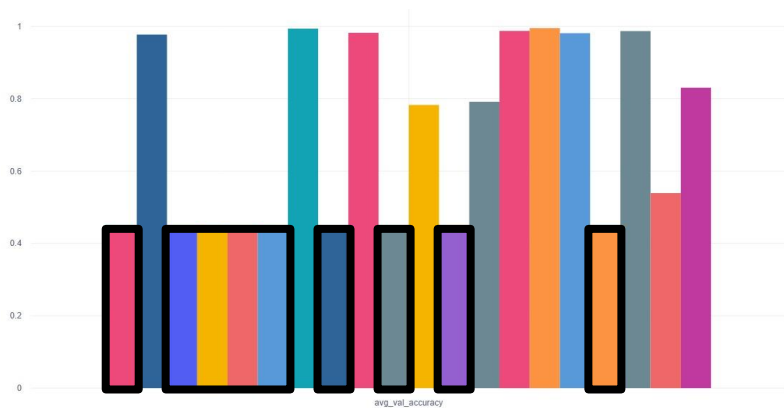
# Convolutional Network



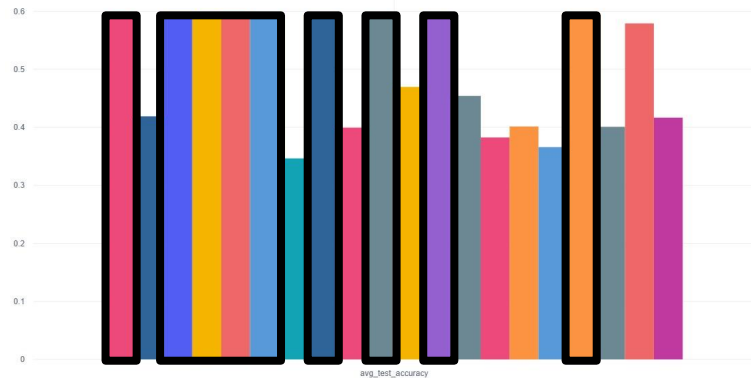
| hyperparameter | type        | range                 | explanation                                   |
|----------------|-------------|-----------------------|-----------------------------------------------|
| units          | integer     | {32, 64, 128, 256}    | Number of output features of the Dense layer. |
| activation     | categorical | {tanh, sigmoid, relu} | Activation function for the output.           |

[https://www.researchgate.net/figure/Schematic-diagram-of-a-basic-convolutional-neural-network-CNN-architecture-26\\_fig1\\_336805909](https://www.researchgate.net/figure/Schematic-diagram-of-a-basic-convolutional-neural-network-CNN-architecture-26_fig1_336805909)[accessed 21 June 2025]

# Convolutional Network

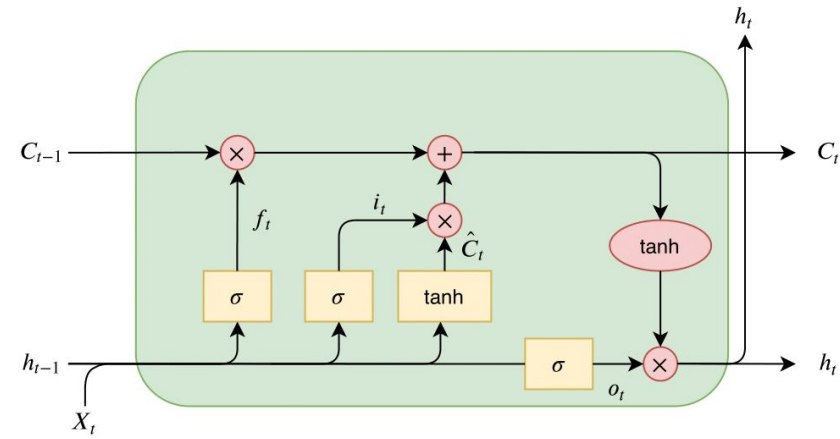


43,67%



59,24%

# Long Short-Term Memory Network

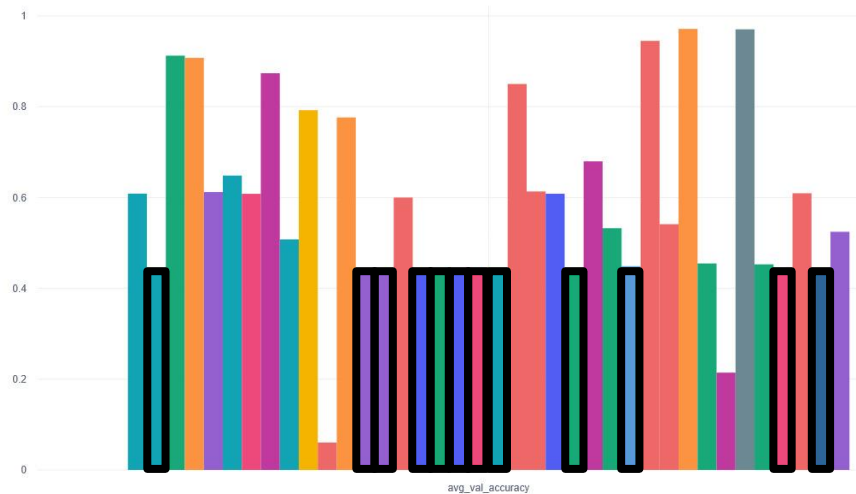


| hyperparameter         | type    | range                    | explanation                               |
|------------------------|---------|--------------------------|-------------------------------------------|
| number of LSTM layers  | integer | $[1, 4] \cap \mathbb{Z}$ | Number of LSTM layers used in the model.  |
| number of Dense layers | integer | $[1, 3] \cap \mathbb{Z}$ | Number of Dense layers used in the model. |

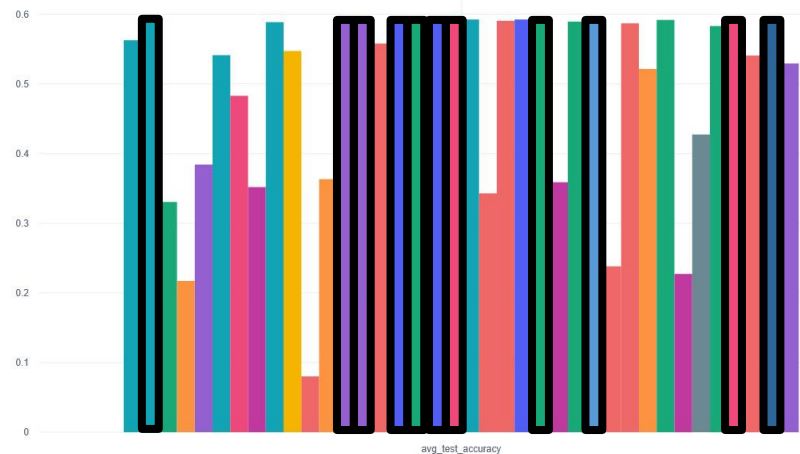
For each LSTM layer, we additionally include the following hyperparameters.

| hyperparameter       | type        | range                             | explanation                                                     |
|----------------------|-------------|-----------------------------------|-----------------------------------------------------------------|
| units                | integer     | $[1, 64] \cap \mathbb{Z}$         | Number of output features of the LSTM layer.                    |
| activation           | categorical | <del>{relu}</del> , tanh, sigmoid | Activation function.                                            |
| recurrent activation | categorical | <del>{relu}</del> , tanh, sigmoid | Activation function for the recurrent step.                     |
| go backwards         | categorical | {true, false}                     | Whether the sequence should be processed forwards or backwards. |

# Long Short-Term Memory Network



**43,67%**



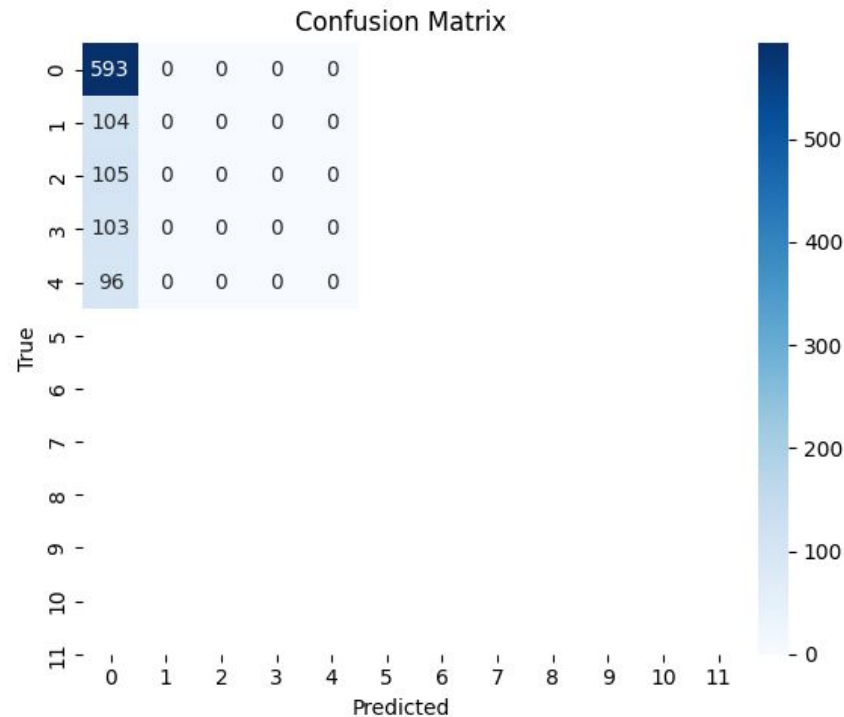
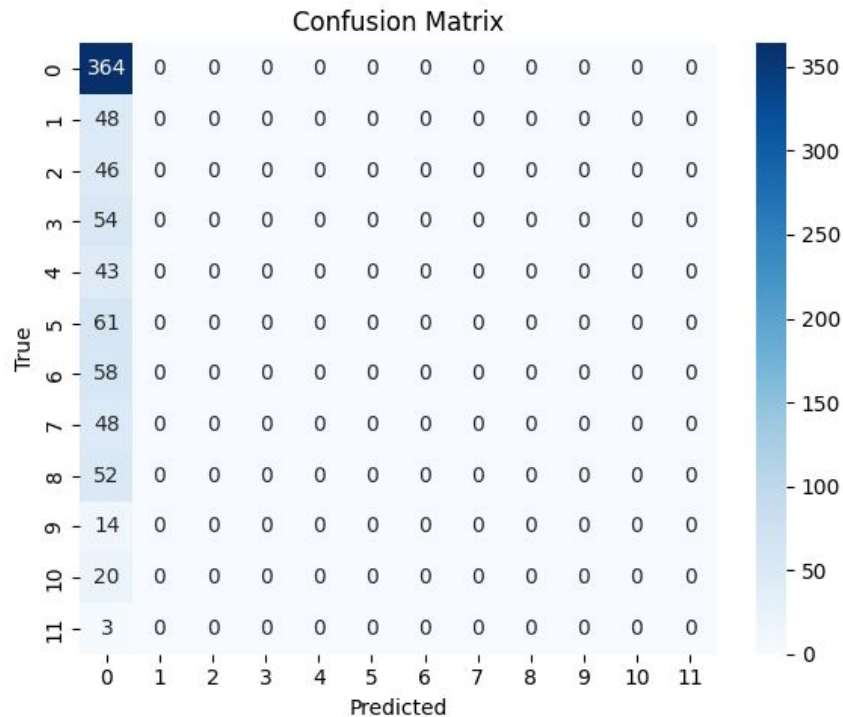
**59,24%**

# Comparison

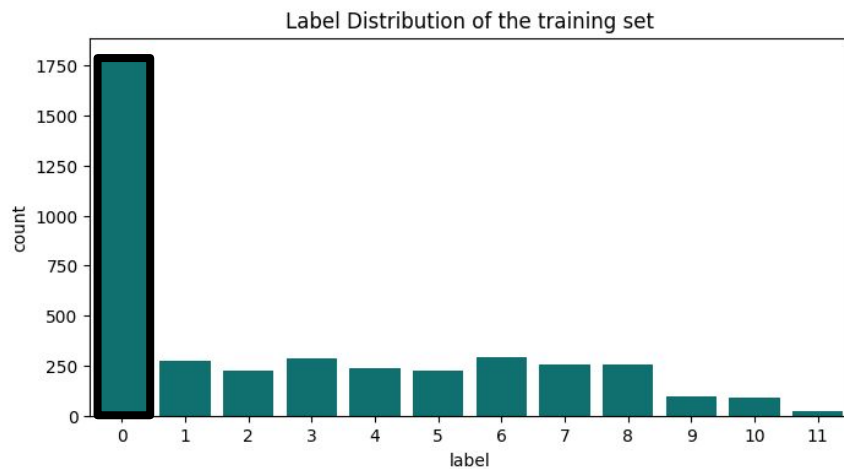
FCN 39,12%

CNN 40,14%

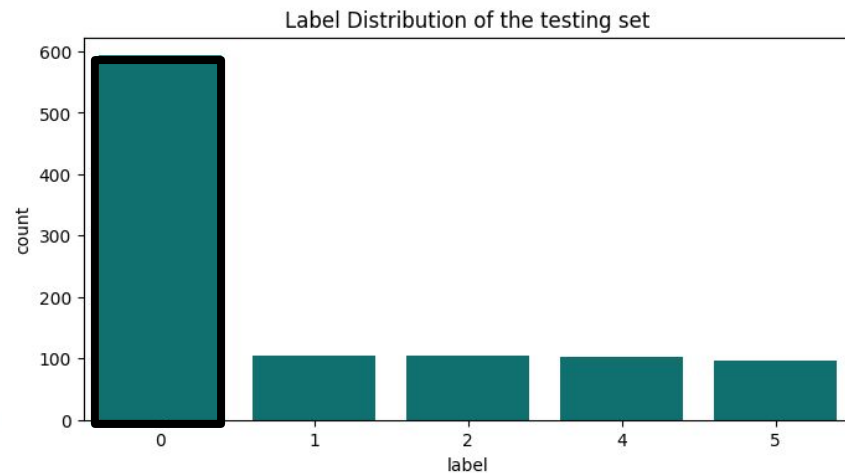
LSTM 52,14%



# Comparison

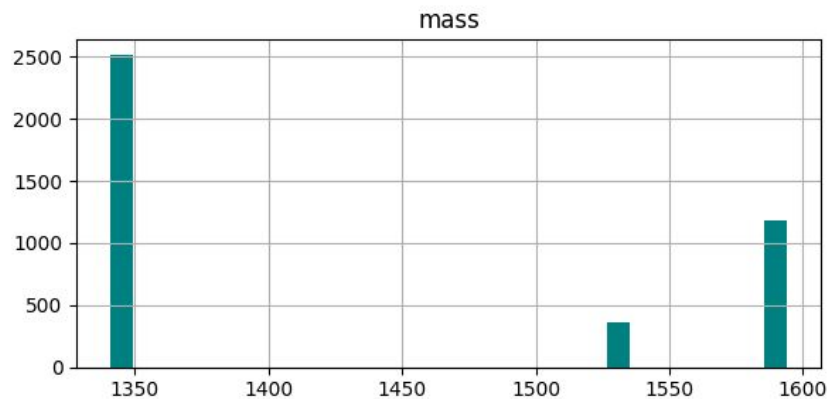


43,67%

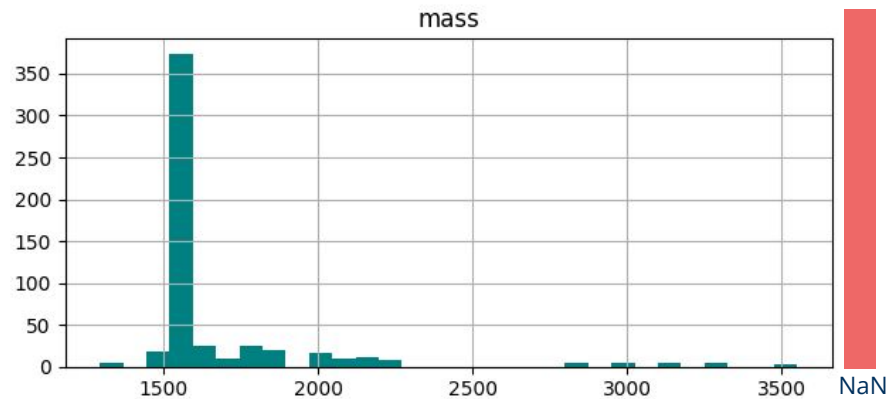


59,24%

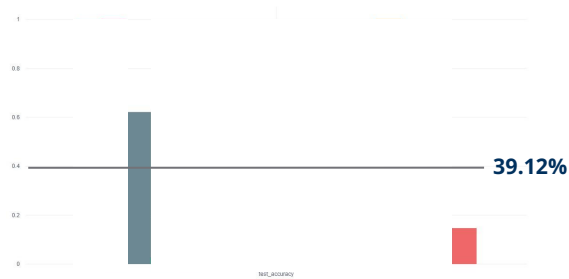
# Analysis with Respect to Metadata - Mass



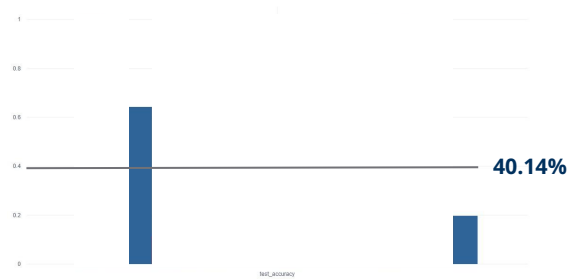
Train



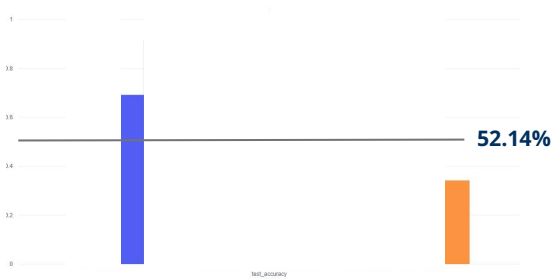
Test



FCN

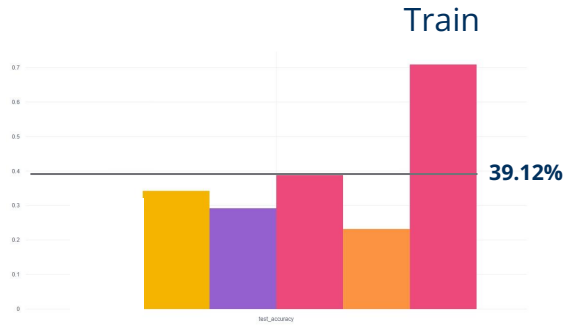
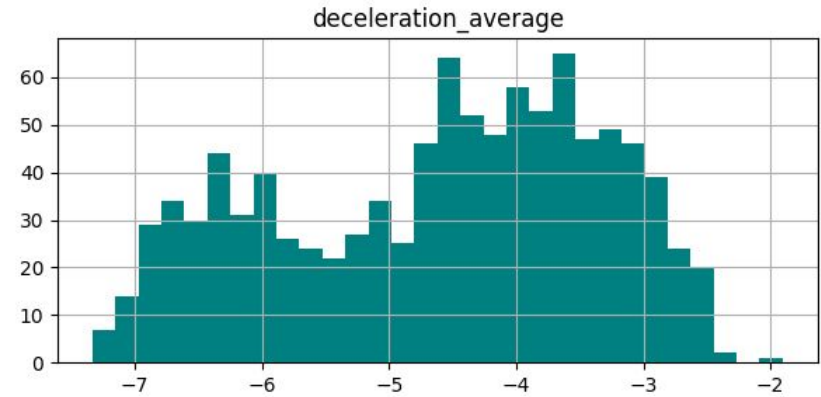
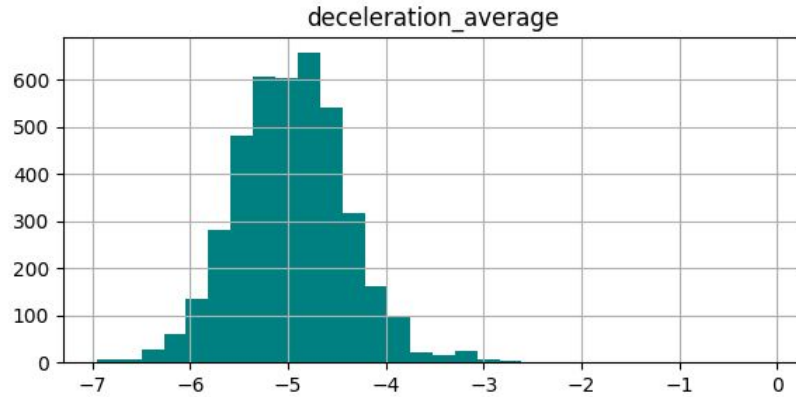


CNN

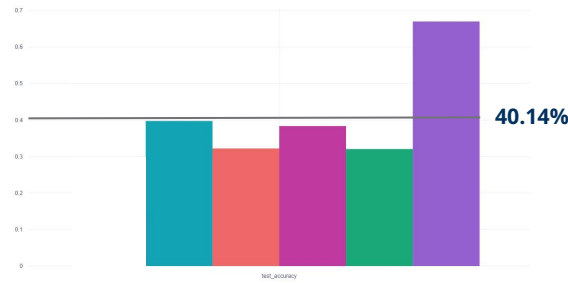


LSTM

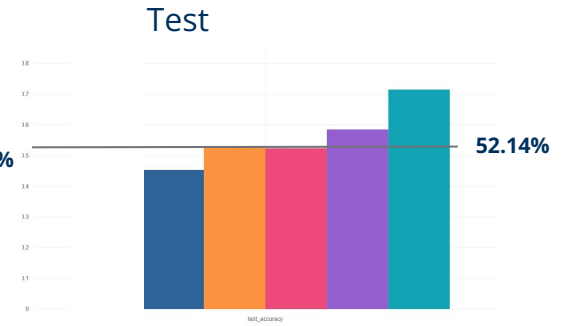
# Analysis with Respect to Metadata - Deceleration Average



FCN



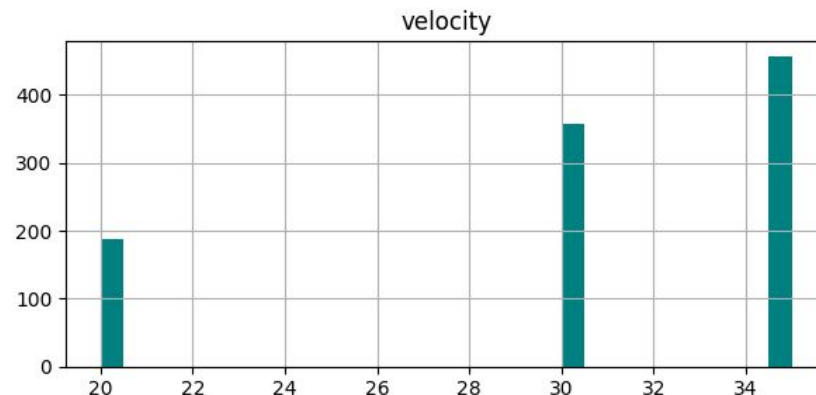
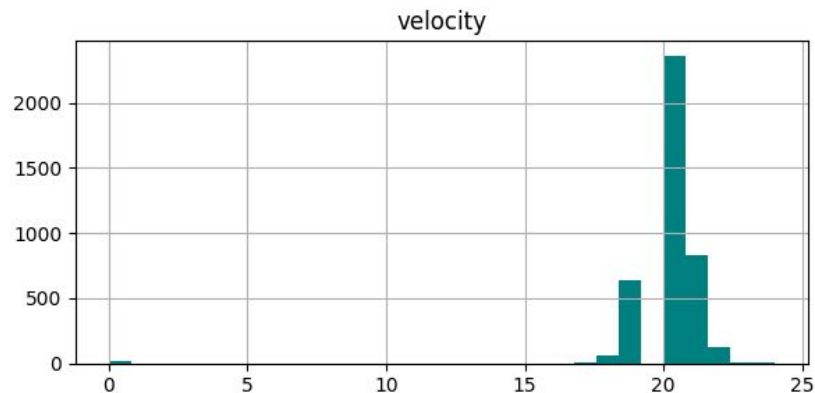
CNN



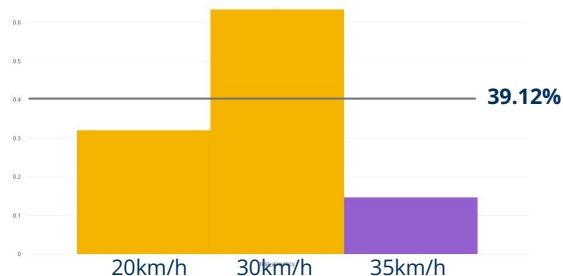
LSTM



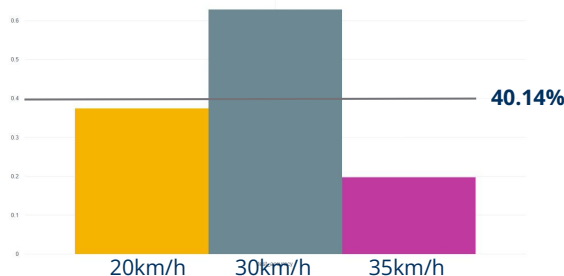
# Analysis with Respect to Metadata - Velocity



Train

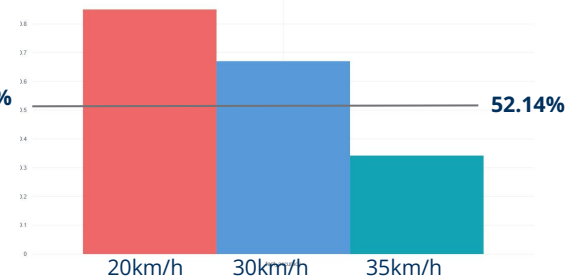


FCN



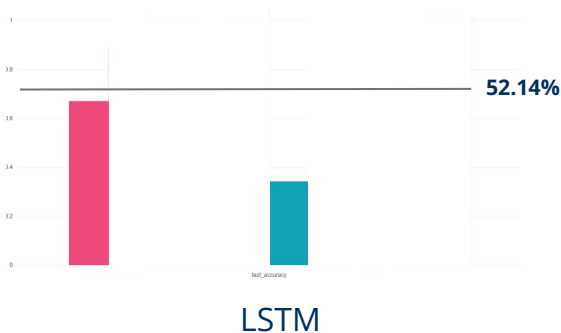
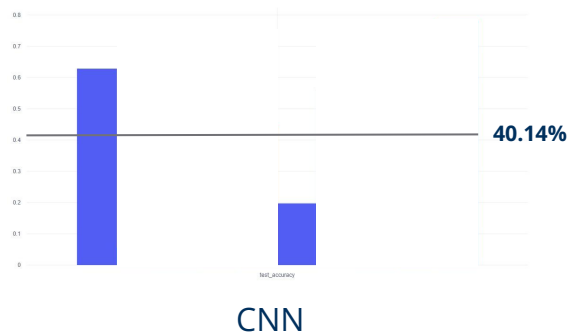
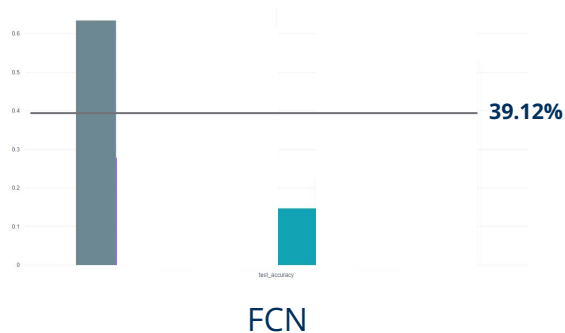
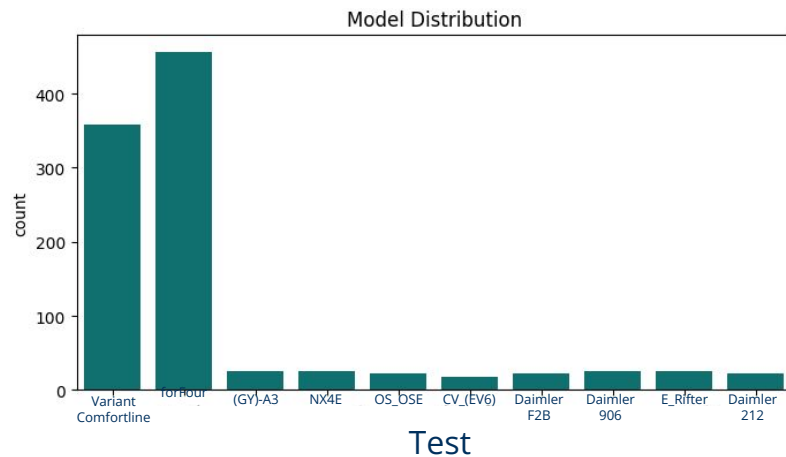
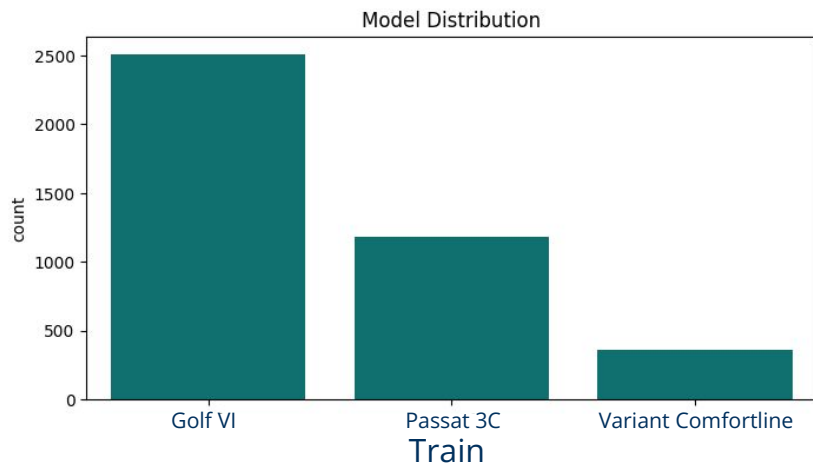
CNN

Test

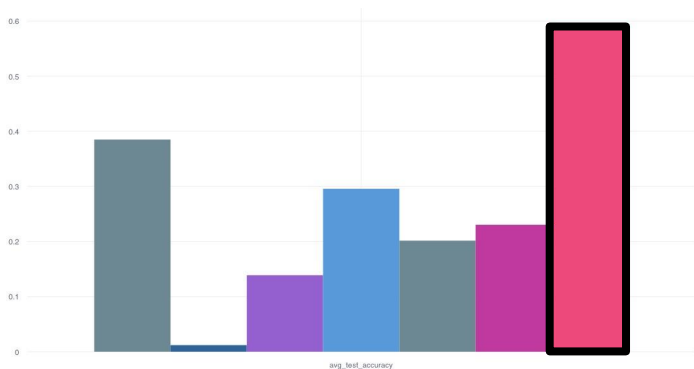


LSTM

# Analysis with Respect to Metadata - Car Model



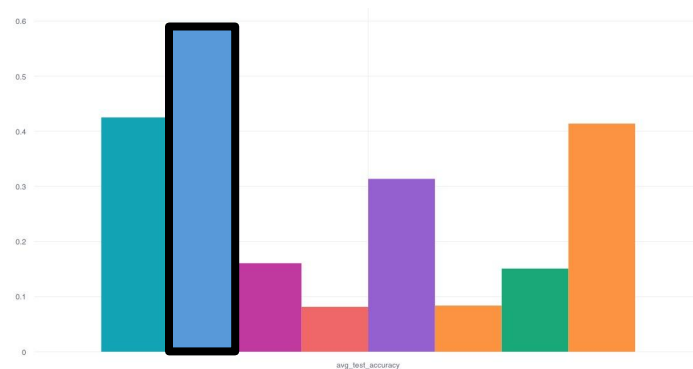
# Data Preprocessing



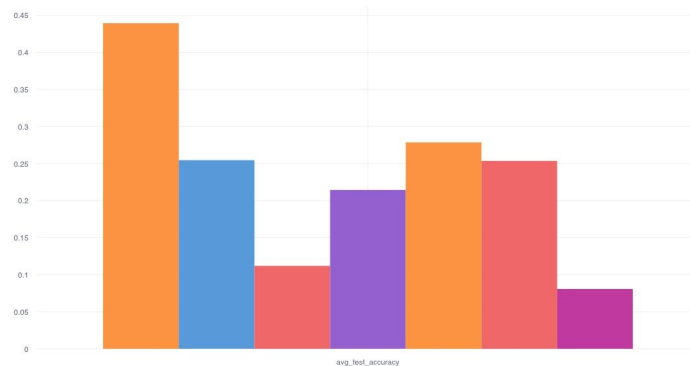
average test acc of FCN

from left to right

1. unprocessed
2. standardised array
3. normalised array
4. standardised row
5. normalised row
6. standardised column
7. normalised column

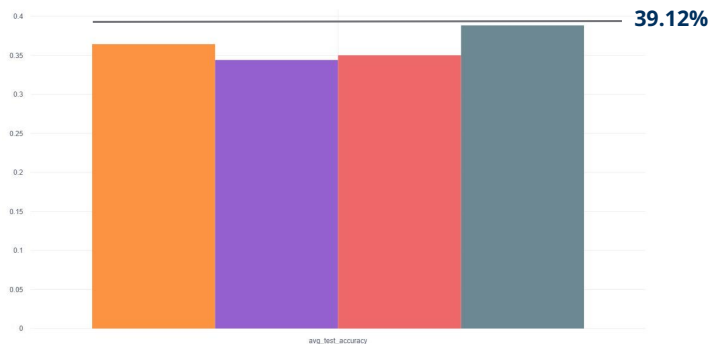


average test acc of CNN



average test acc of LSTM

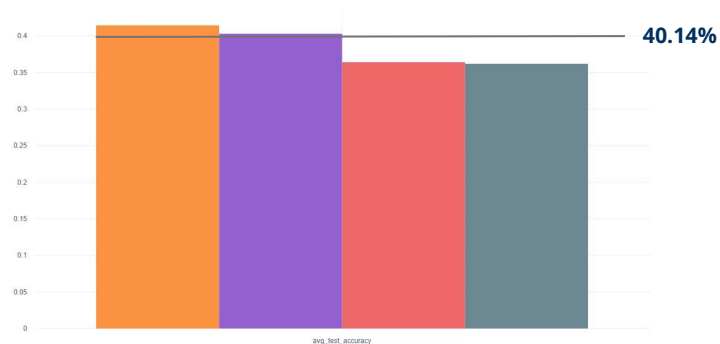
# Data Augmentation



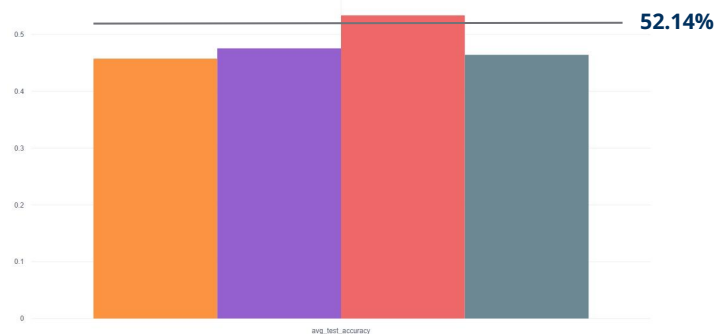
test accuracy of the augmentation methods for FCN

augmentation methods:

- AmplitudeScaling
- NoiseAdding
- ZoomIn
- Flip

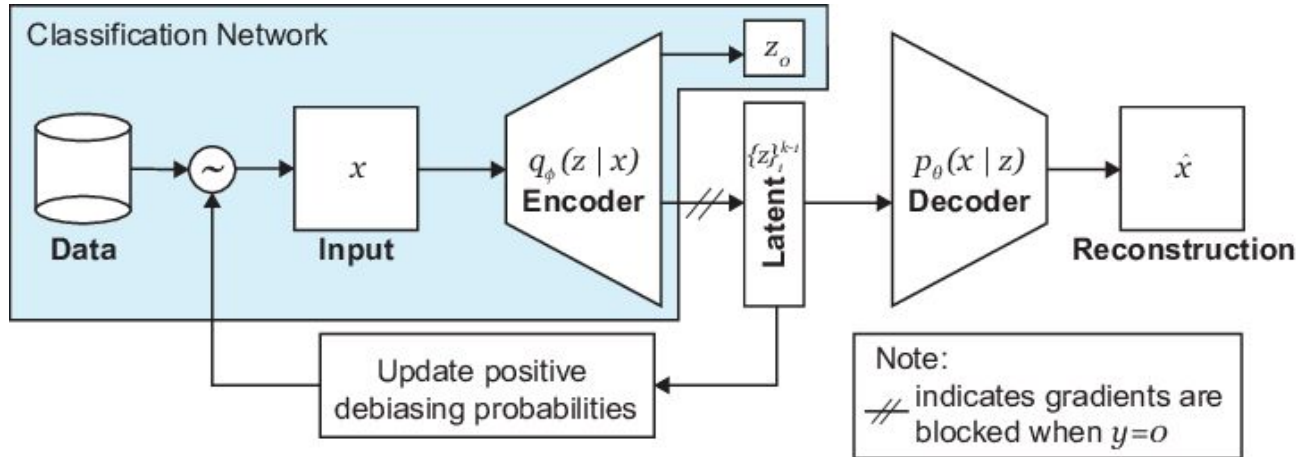


test accuracy of the augmentation methods for CNN



test accuracy of the augmentation methods for LSTM

# Debiasing Variational Autoencoder (DB-VAE)



# Debiasing Variational Autoencoder (DB-VAE)

$$\mathcal{L}_{\text{latent}}(x) = \frac{1}{2} \sum_{j=1}^d (\exp(\log \sigma_j^2) + \mu_j^2 - 1 - \log \sigma_j^2) \quad (1)$$

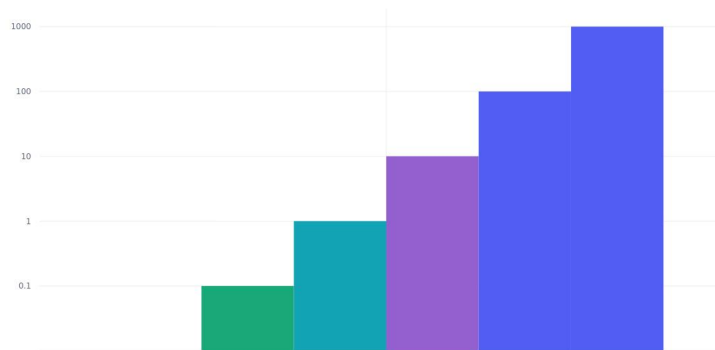
$$\mathcal{L}_{\text{reconstruction}}(x, \hat{x}) = \frac{1}{768} \sum_{i=1}^6 \sum_{k=1}^{128} |x_{i,k} - \hat{x}_{i,k}| \quad (2)$$

$$\mathcal{L}_{\text{VAE}} = w_{\text{KL}} \cdot \mathcal{L}_{\text{latent}} + \mathcal{L}_{\text{reconstruction}} \quad (3)$$

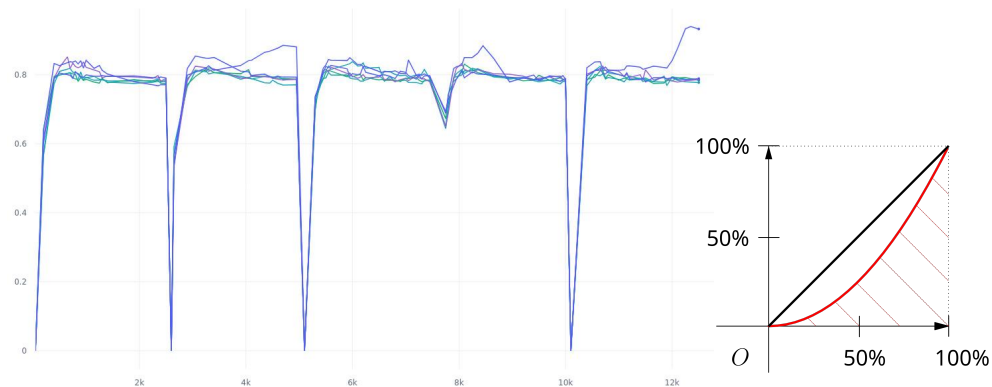
$$\mathcal{L}_{\text{classification}}(y, \hat{y}) = -\log \hat{y}_y \quad (4)$$

$$\mathcal{L}_{\text{total}} = \mathbb{E} [\mathcal{L}_{\text{classification}} + w_{\text{VAE}} \cdot \mathcal{L}_{\text{VAE}}] \quad (5)$$

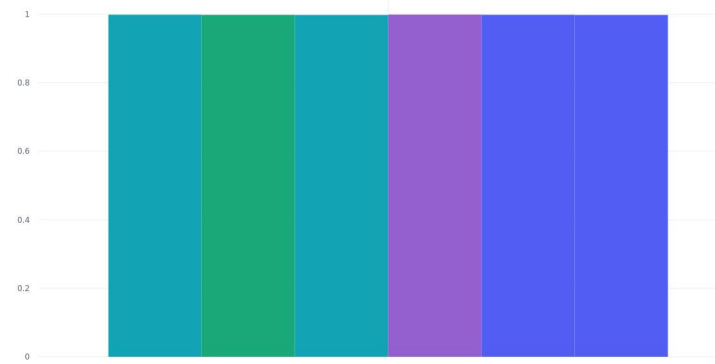
# Evaluating the DB-VAE (Part 1)



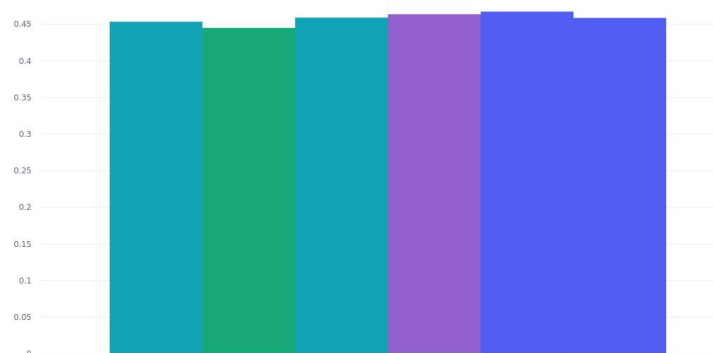
VAE-loss weight



Gini inequality index of resampling distribution

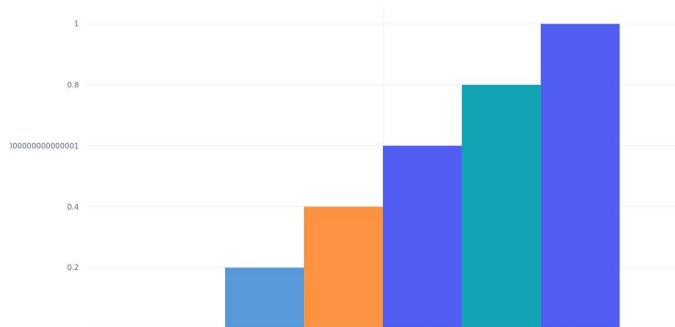


Average Validation Accuracy

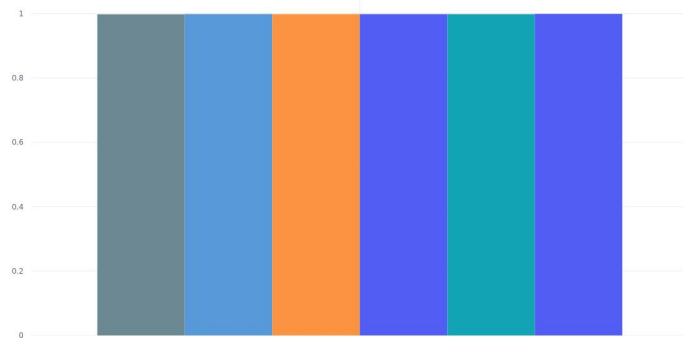


Average Test Accuracy

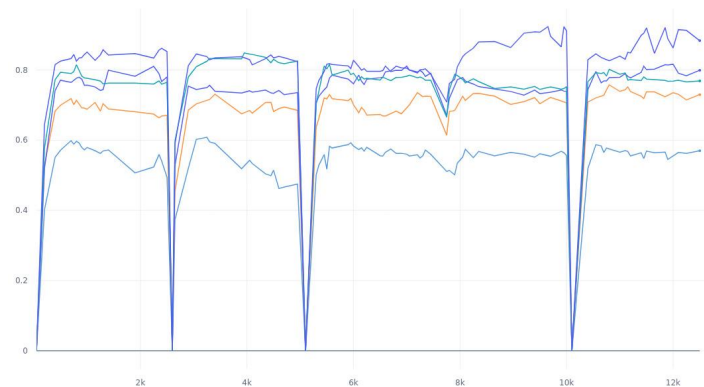
# Evaluating the DB-VAE (Part 2)



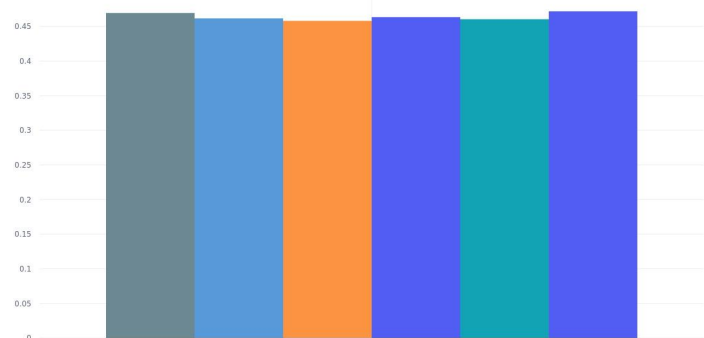
Debiasing Aggressiveness (Alpha)



Average Validation Accuracy



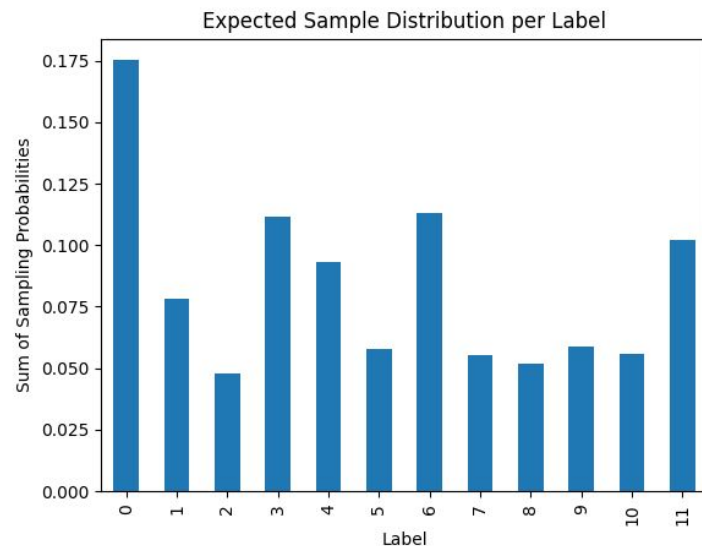
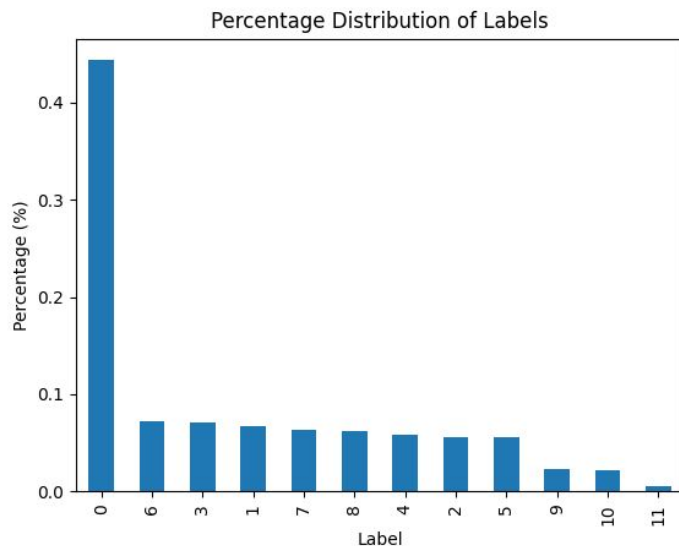
Gini inequality index of resampling distribution



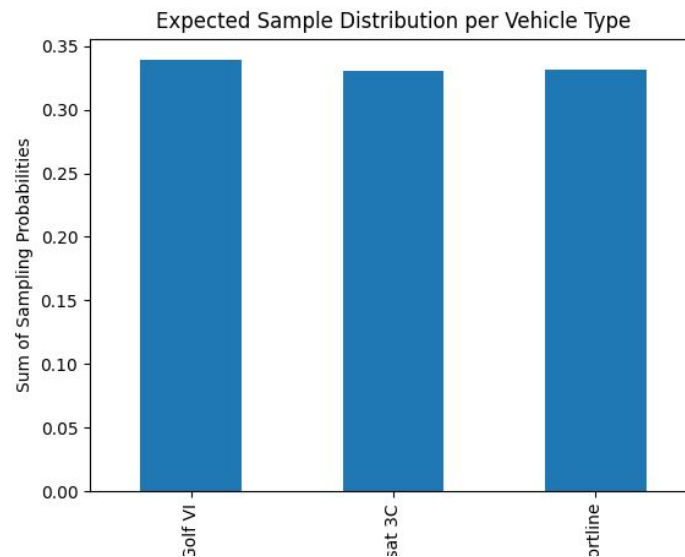
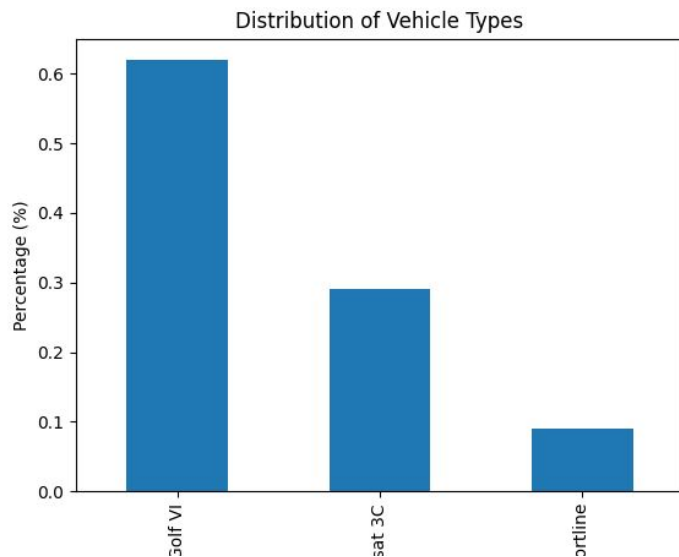
Average Test Accuracy



# Effect of debiasing (adaptive resampling) in the last epoch (Part 1)



# Effect of debiasing (adaptive resampling) in the last epoch (Part 2)



# Summary

- **FCN, CNN and LSTM about equally good, slight advantages for the LSTM**
- **very good results on the validation data**
- **poor performance for the test data**
- **problem: just guessing class 0 all the time**
- **debiasing did not work very well**
  - preprocessing led to worse result
  - augmentation worked better
  - with DB-VAE unchanged results

# Questions?

# Image references

- Long Short-Term Memory Layer Architecture. Available from:  
<https://medium.com/@saba99/long-short-term-memory-lstm-fffc5eaebfdc> [accessed 21 Jun 2025]
- Uncovering and Mitigating Algorithmic Bias through Learned Latent Structure - Scientific Figure on ResearchGate. Available from:  
[https://www.researchgate.net/figure/Debiasing-Variational-Autoencoder-Architecture-of-the-semi-supervised-DB-VAE-for-binary\\_fig2\\_334381622](https://www.researchgate.net/figure/Debiasing-Variational-Autoencoder-Architecture-of-the-semi-supervised-DB-VAE-for-binary_fig2_334381622) [accessed 21 Jun 2025]
- Gini Index Definition. Von Aleks-ger - Eigenes Werk, CC BY-SA 3.0,  
<https://commons.wikimedia.org/w/index.php?curid=32716745>